

Skills

Python

NumPy, Pandas, Scikit-learn, Matplotlib, SciPy

Python machine learning

Regression, Decision Trees, Controller

3D CAD modelling

Autodesk Inventor, SolidWorks, Autodesk REVIT, IESVE

Data analytics and visualization

HTML/JavaScript, Plotly, D3

Engineering & Human-centered design

HMI, Sustainable design

Research Interests

Scientific and Social Aspects of Sustainability, Distributed Energy Resources, Design, Renewable Energy Generation, Robotics & Controls, Energy Storage

Languages

English

Mandarin

Malay

Publications

Decision Tree Variations and Online Tuning for Real-Time Control of a Building in a Two-Stage Management Strategy

DOI: [10.3390/en17112730](https://doi.org/10.3390/en17112730)

References

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Centre national de la recherche scientifique (CNRS)

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Summary

Research Engineer bridging sustainable thermal design and data science. Currently developing low-cost cold chain solutions via 3D prototyping, PCM experimentation, and Python analytics. Experienced in ML optimization, experimental design, and academic publishing across energy systems and engineering research.

Education

Newcastle University

Engineering

MSc Energy & Sustainability • Distinction

Singapore • 2018-2019

Newcastle University

Engineering

BSc MDME • Hons (2:1)

Singapore (SIT) • 2016-2018

Mechanical Design & Manufacturing Engineering

Experience

Republic Polytechnic

Research Staff

Singapore

July 2025 - Present

Designed and iterated cold box prototypes via CAD and FDM 3D printing for rapid design-test cycles. Conducted material trade-offs to select sustainable, low-cost insulation and structural materials.

Designed and executed DOE (Design of Experiments) frameworks for shelf-life stability and PCM thermal characterization. Developed SOPs for all experimental protocols, ensuring reproducibility and lab safety.

Built HTML/JavaScript and Python dashboards for data analysis/visualization.

Supervised several Applied Science FYP teams for shelf-life studies (microbial, vitamin A&C, texture analysis, moisture analysis and water activity). Supervised Engineering FYP students for the design, material and in-house PCM testing for cold box prototype.

CNRS@CREATE

Research Associate

Singapore

Jan 2023 - Sep 2023

Conversion of a MATLAB optimization program into Python for the purpose of replacing an optimization controller of a grid connected distributed energy system into an adaptive data-driven (machine learning) lightweight compact linear tree controller

Surbana Jurong-NTU Corporate Lab

Research Associate

Singapore

Apr 2022 - Sep 2022

Created a Python program for the generation and prediction of efficiency values for cooling systems using python machine learning regression algorithm

Created a Python program for the optimization of chiller systems sizing and daily operations using python

NTU Institute of Science & Technology for Humanity

Research Associate

Singapore

Nov 2020 - Apr 2022

Using python/JavaScript for data collection, analysis & generating data visualization of a network of NISTH fellow's publications in relation to the UN Sustainable development Goals on an online platform.

Building & design of NISTH 2021 Annual web report

Assist in report writing & idea generation for new research topics related to science & technology with humanities angles

Supervise FYP undergraduate on their social robot for elderly to promote inter-generational connectivity